

EC0402

ASSIGNMENT NO#1

STUDENT NAME:

STUDENT ID:

BY VU BSCS

Case 01: Plastic and Toys Industry

Requirement #01:

We are given the **demand** and **supply** functions for toys:

- **Demand Function:** $Q_d = 2100 - 3P$

Supply Function: $Q_s = 1700 + 10P$

We need to calculate the quantity demanded and supplied at two different price points, **Rs. 30** and **Rs. 110**.

Step 1: Calculate Quantity Demanded and Supplied at Rs. 30

1. **Quantity Demanded** at **P = 30**:

Using the demand function:

$$Q_d = 2100 - 3P$$

$$Q_d = 2100 - 3(30) = 2100 - 90 = 2010$$

So, the **quantity demanded** when the price is Rs. 30 is **2010 toys**.

2. **Quantity Supplied** at **P = 30**:

Using the supply function:

$$Q_s = 1700 + 10P$$

$$Q_s = 1700 + 10(30) = 1700 + 300 = 2000$$

So, the **quantity supplied** when the price is Rs. 30 is **2000 toys**.

Step 2: Calculate Quantity Demanded and Supplied at Rs. 110

1. Quantity Demanded at $P = 110$:

Using the demand function:

$$Q_d = 2100 - 3P$$

$$Q_d = 2100 - 3(110) = 2100 - 330 = 1770$$

So, the **quantity demanded** when the price is Rs. 110 is **1770 toys**.

2. Quantity Supplied at $P = 110$:

Using the supply function:

$$Q_s = 1700 + 10P$$

$$Q_s = 1700 + 10(110) = 1700 + 1100 = 2800$$

So, the **quantity supplied** when the price is Rs. 110 is **2800 toys**.

Requirement #01 Summary (Table-1):

Price (Rs.)	Quantity Demanded (QD)	Quantity Supplied (QS)	Market Situation
Rs. 30	2010	2000	Shortage
Rs. 110	1770	2800	Surplus

Market Conditions Interpretation:

- **At Rs. 30**, the quantity demanded (2010 toys) is greater than the quantity supplied (2000 toys), indicating a **shortage** of toys.
- **At Rs. 110**, the quantity demanded (1770 toys) is less than the quantity supplied (2800 toys), indicating a **surplus** of toys.

Requirement #02:

We are asked to calculate the **elasticity of demand** for toys when the price changes from **Rs. 20 to Rs. 40**.

We are given the following information from **Table-2**:

Price (Rs.)	Quantity Demanded	Quantity Supplied
Rs. 20	2040	1900
Rs. 40	1980	2100

We will calculate the **Price Elasticity of Demand (PED)** using the following formula:

$$\text{PED} = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in price}}$$

The **percentage change in quantity demanded** and the **percentage change in price** can be calculated as follows:

Step 1: Calculate the change in quantity demanded and price.

- **Initial price (P1) = Rs. 20, New price (P2) = Rs. 40**
- **Initial quantity demanded (QD1) = 2040, New quantity demanded (QD2) = 1980**

Change in quantity demanded:

$$\Delta QD = QD2 - QD1 = 1980 - 2040 = -60$$

Change in price:

$$\Delta P = P2 - P1 = 40 - 20 = 20$$

Step 2: Calculate the percentage changes.

- **Percentage change in quantity demanded:**

$$\frac{\Delta QD}{QD1} \times 100 = \frac{-60}{2040} \times 100 \approx -2.94\%$$

Percentage change in price:

$$\frac{\Delta P}{P1} \times 100 = \frac{20}{20} \times 100 = 100\%$$

Step 3: Calculate the price elasticity of demand.

$$\text{PED} = \frac{-2.94\%}{100\%} = -0.0294$$

Interpretation:

Since the **absolute value of PED** is **less than 1**, we can conclude that the demand for toys is **inelastic** over the price range from Rs. 20 to Rs. 40. This means that the quantity demanded is not very responsive to the price change. A 1% increase in price results in less than a 1% decrease in quantity demanded

Final Answers:

Requirement #01:

- At **Rs. 30:**

Quantity demanded = 2010 toys

Quantity Supplied = 2000 toys

Market Situation = **Shortage**

- At **Rs. 110:**

Quantity demanded = 1770 toys

Quantity Supplied = 2800 toys

Market Situation = **Surplus**

Requirement #02:

- **Price Elasticity of Demand** when the price increases from Rs. 20 to Rs. 40 is **-0.0294** (inelastic demand).

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